1. Purpose of the Digital Twin

The Digital Twin provides a **computational replica** of Butembo's road accident dynamics, enabling:

- risk prediction,
- spatial propagation analysis,
- temporal forecasting,
- policy and infrastructure scenario testing.

2. Methodological Mapping

Folder	Scientific role
preprocessing	Data cleaning & feature engineering
graph	Spatial structure reconstruction
models/boosting	Nonlinear tabular modelling
models/lstm	Temporal dynamics
models/gnn	Spatial propagation
ensemble	Model fusion & ablation
evaluation	Statistical validation
simulation	Digital Twin engine
api	Deployment
dashboard	Visualization

3. File-by-File Explanation

preprocessing/preprocessing.py

- Normalizes time formats
- Harmonizes categorical variables
- Removes PII (Personally Identifiable Information)
- Generates ML-ready matrices

models/lstm/build_hourly_series.py

- Builds the 24-hour accident profile
- Prepares LSTM input

graph/build_graph.py

- Builds adjacency structure without GPS
- Uses co-occurrence & textual logic

graph/node_features.py
- Computes node descriptors:
  - accident frequency
  - fatality rate
  - temporal patterns

models/boosting/*
- Train CatBoost, XGBoost, LightGBM
  - Output probabilistic predictions

models/lstm/train_lstm.py
- Learns intra-day accident rhythms

models/gnn/train_gnn.py
- Learns spatial risk propagation

ensemble/ablation.py
- Quantifies contribution of each model

ensemble/fusion.py
- Computes hybrid probability:

$$\hat{p} = \propto p_{boost} + \beta p_{lstm} + \gamma p_{gnn}$$

evaluation/*
- Computes F1, AUC
- Calibration error
- Comparison tables

simulation/digital_twin.py
- 24h risk simulation
- Intervention testing

api/main.py
- Serves predictions & simulations

dashboard/app.py
- Interactive visualization

4. Execution Dependencies

requirements.txt

All libraries required to reproduce the experiments and simulations.

5. Execution Logic (Dependency Graph)

1. preprocessing → creates ML inputs
2. graph → creates spatial structure
3. models → produce predictions
4. ensemble → fuses predictions
5. evaluation → validates performance
6. simulation → generates Digital Twin
7. api and dashboard → expose results

6. Intended Users

- Researchers
- Urban planners
- Policy makers
- Academic evaluators

7. Conclusion

This Digital Twin is designed to be:

- modular,
- reproducible,
- defensible in academic contexts,
- adaptable to other African cities.

8. Appendices

The FastAPI API exposes the Butembo Digital Twin through two main endpoints:

/baseline and /scenarios

8.1. What does /baseline mean?

/baseline returns the reference simulation (baseline scenario) of the Digital Twin.
It represents:

- the current state of the road network,

- without any intervention,
- based solely on:
    - historical data,
    - the hybrid model (Boosting + LSTM + GNN),
    - hourly temporal dynamics.
- The /baseline response contains a JSON object with the following structure:

```
{
  "0": [p0, p1, p2, ..., pN],
  "1": [p0, p1, p2, ..., pN],
  ...
  "23": [p0, p1, p2, ..., pN]
}
```

Where:

- the key "h" (0 – 23) represents the hour of the day,
- the value is a list of predicted risks,
- with one risk value per road segment.

Scientific interpretation

/baseline answers the question:
"How does accident risk evolve over the course of a normal day, without any intervention?"
It serves as:

- the mandatory reference point
- for any impact analysis or public policy decision.

8.2. What does /scenarios mean?

/scenarios returns the counterfactual simulations of the Digital Twin.
Each scenario modifies one or more system parameters, such as:

- driver behavior,
- speed regulation,
- road segment closures,
- infrastructure improvements, etc.

scenarios response returns a **JSON object structured by scenario**, including the simulated outcomes for each configuration:

```
{
  "speed_limit_20pct": {
    "0": [...],
    "1": [...],
    ...
    "23": [...]
  },
  "targeted_closure": {
    "0": [...],
    ...
  }
}
```


/scenario responds to the question:
"What would happen if a specific policy or intervention were applied?"

This is the **decision-making core** of the Digital Twin:

- evaluation before real-world deployment,
- comparison between baseline and intervention scenarios,
- support for planning and decision-making.

"The /baseline endpoint provides the reference risk simulation of the Digital Twin under current conditions, while the /scenarios endpoint exposes counterfactual simulations that quantify the impact of targeted interventions over a full 24-hour cycle."

Endpoint	Role	Key question
/baseline	Reference	*What happens without any action?*
/scenarios	Contrefactual	*What happens if action/intervention is taken?*

/baseline shows the risk as it is, while **/scenarios** shows how the risk could change if an intervention is implemented.

- "In this study, an intervention refers to a virtual modification of traffic or infrastructure conditions — such as speed reduction or targeted road closures — applied within the digital twin to assess its potential impact on accident risk before real-world deployment."

- The "intervention" is not a real-world action, but a simulated policy or measure (such as speed regulation, road closure, or enforcement) that is tested virtually within the Digital Twin.

- **Where are interventions defined concretely?**
  - They are defined directly in the code: simulation/digital_twin.py

- **For example:**
  - **Intervention 1 — Speed Reduction**

```
simulate_day(intervention={"speed_reduction": 0.8})
```

Meaning:

- −20% average speed,
- therefore −20% risk, all else being equal.

Real-world interpretation:

- speed limits,
- speed bumps,
- strengthened police enforcement

  - **Intervention 2 — Targeted closure of road segments**

```
simulate_day(intervention={"closure_nodes": [0, 1, 2, 3, 4]})
```

Meaning:

- temporary closure of high-risk road segments.

Real-world interpretation:

- road works,
- detours,
- pedestrianization,
- temporary motorcycle bans.

Thus, concretely, what kinds of interventions are we talking about?

In your project, interventions are generic but realistic scenarios, for example:

Simulated Intervention sin this project	Real Meaning
speed_reduction = 0.8	Speed limitation
closure_nodes	Road closure / diversion
(extensible)	Improved lighting, road surface, and police presence